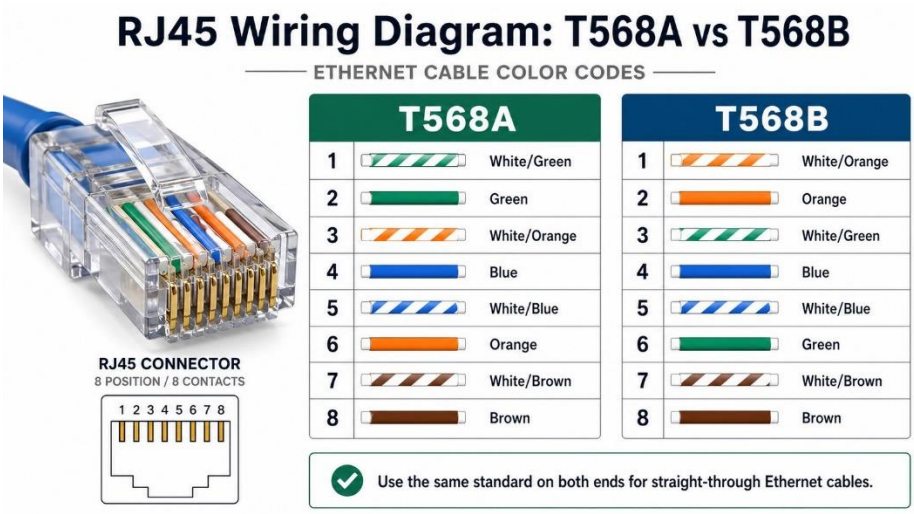
Ethernet standard and specification:-

Tasks:-

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Ans.:-



The RJ-45 connector is the standard connector used for Ethernet (LAN) cables to connect computers, switches, routers, and other network devices. It has 8 pins and uses 4 twisted pairs (8 wires).

Pin Layout:-

The pins are numbered 1 to 8 from left to right when looking at the front of the connector with the gold contacts facing you and the clip facing away.

T568A Standard:-

Wire order:

1. White/Green
2. Green
3. White/Orange
4. Blue
5. White/Blue

6. Orange
7. White/Brown
8. Brown

Used in: Some government and residential installations.

T568B Standard:-

Wire order:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Used in: Most offices, schools, businesses, and modern Ethernet networks.

Main Difference :-

Only the green and orange wire pairs are swapped.

- **T568A:** Green pair → Pins 1–2, Orange pair → Pins 3–6
- **T568B:** Orange pair → Pins 1–2, Green pair → Pins 3–6

The blue and brown pairs remain in the same positions in both standards.

2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ans.:-

Specification	10BASE-T	100BASE-TX (Fast Ethernet)	1000BASE-T (Gigabit Ethernet)
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable Type	Cat3, Cat5, Cat5e, Cat6	Cat5 or higher (Cat5e/Cat6 recommended)	Cat5e or Cat6
Maximum Cable Length	100 meters (328 ft)	100 meters (328 ft)	100 meters (328 ft)
Connector Type	RJ-45	RJ-45	RJ-45
Twisted Pairs Used	2 pairs	2 pairs	4 pairs
IEEE Standard	IEEE 802.3i	IEEE 802.3u	IEEE 802.3ab
Typical Use	Older LANs, legacy devices	Small office and home networks	Modern offices, gaming, streaming, and enterprise networks

**3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.**

Ans.:-

Scenario:-

Suppose you need to transfer large files between two laptops or create a small local network without using a switch, router, or Wi-Fi. In this case, the two laptops are connected directly using an Ethernet cable.

To connect two laptops directly using Ethernet, you would traditionally use a crossover cable.

Reason:-

A crossover cable swaps the transmit (TX) and receive (RX) wire pairs so that:-

- The TX pins of Laptop 1 connect to the RX pins of Laptop 2.
- The RX pins of Laptop 1 connect to the TX pins of Laptop 2.

Since both laptops are the same type of device (end devices), their transmit and receive pins are wired the same way. Without crossing the pairs, both devices would try to transmit on the same pins and listen on the same pins, preventing communication.

Feature	**Straight-Through Cable**	**Crossover Cable**
Wiring Standard	Same on both ends (T568A-A or T568B-B)	Different on each end (T568A-B)
TX/RX Pairs	Not crossed	Crossed
Used For	Different devices (PC - Switch, Router - Switch)	Similar devices (Laptop - Laptop, PC - PC, Switch - Switch on older hardware)
Common Today	Yes	Mostly for older devices

Most modern laptops have Auto MDI-X technology. Auto MDI-X automatically detects the connection type and adjusts the transmit and receive pairs internally. Because of this, either a straight-through or crossover cable usually works on modern laptops.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Ans.:-

Device	Location	Likely Ethernet Standard	Reason
Desktop Computer	College Computer Lab	1000BASE-T (1 Gbps)	Most modern desktop PCs have Gigabit Ethernet for fast file transfers and internet access.
Wi-Fi Router	Home	1000BASE-T (1 Gbps)	Modern routers usually provide Gigabit LAN ports for high-speed wired connections.
Network Printer	College Office/Library	100BASE-TX (100 Mbps)	Many network printers use Fast Ethernet because printing requires less bandwidth than computers.